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DESIGN AND CONFIGURATION OF AN INTERNETWORK IN THE LAB USING SWITCH, ROUTER AND ETHERNET CABLES

AIM:

To design and configure an internetwork in the laboratory using switches, a router and Ethernet cables, assign IP addresses to the connected computers, and verify communication between different networks.

ALGORITHM:

1. Collect the required devices, including computers, switches, a router and Ethernet cables.
2. Connect the computers to their respective switches using Ethernet cables.
3. Connect the switches to the router using Ethernet cables.
4. Assign static IP addresses and subnet masks to all computers.
5. Configure the router interfaces with suitable IP addresses for each network.
6. Enable the router interfaces using the no shutdown command.
7. Configure the default gateway on each computer using the corresponding router interface IP address.
8. Verify the IP configuration of each computer.
9. Use the ping command to test connectivity between computers in the same network.
10. Test communication between computers belonging to different networks.
11. Observe the packet transmission and routing process.
12. Record the results and verify successful communication between the connected networks.

IP ADDRESS CONFIGURATION:

Network 1:

PC1 IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Router Interface G0/0: 192.168.1.1

Network 2:

PC2 IP Address: 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

Router Interface G0/1: 192.168.2.1

ROUTER CONFIGURATION:

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.1.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

end

COMMAND:

ping 192.168.2.2

OUTPUT:

When the computers, switches and router are configured correctly, the source computer can communicate with the destination computer in the other network.

Sample output:

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=2ms TTL=127

Reply from 192.168.2.2: bytes=32 time=1ms TTL=127

Ping statistics:

Packets: Sent = 4, Received = 4, Lost = 0

RESULT:

Thus, an internetwork was designed and configured using switches, a router and Ethernet cables. IP addresses were assigned to the connected computers and router interfaces, and communication between different networks was verified using the ping command.